Prompt Library and Verification Report

Capstone 14: AI Accessibility Study Aids | NCERT Class 10 Science, Ch. 11, Section 11.1 | Divyalakshmi.G

PART A: PROMPT LIBRARY

Each prompt gives the tool a role, rules and an output format. The same prompt and the same text were used in all three tools so the comparison is fair. Fill the Date column from your own chat history.

ID	Purpose	Tools	Result	Date
P1	Simplify the NCERT section (text)	ChatGPT, Gemini, NotebookLM	Outputs saved and checked. All three had errors.	
P2	Audio narration script from corrected text	ChatGPT, Gemini, NotebookLM	All spoke numbers correctly. NotebookLM needed 2 attempts.	
P3	Mind-map outline from corrected text	ChatGPT, Gemini, NotebookLM	Structure good; each tool dropped some facts.	
P3b	Design the final mind-map picture	Image / design tool (add name)	Version 1 had 4 problems, version 2 fixed 3.	
A1	Voice generation of final script	ElevenLabs (free tier), model elevenlabs-v4, style warmly	149.9 s audio, no misreads	19 Oct 2026

P1: Simplified text

You are an accessibility specialist helping students who find dense text hard to read (dyslexia, language difficulty). Simplify the section below for a Class 10 student. Rules: sentences under 15 words; explain each technical term in plain words the first time; keep all formulas and units exactly as in the original; do not add facts that are not in the text; use short paragraphs and a bulleted key-terms list at the end. Text: [NCERT Section 11.1 pasted here]

What I learned: the "keep formulas exactly" rule was not enough. Tools still flattened exponents and added words. Pasting a placeholder or only a topic name gives a made-from-memory answer, so the real NCERT text must always be included.

P2: Audio narration script

You are an accessibility specialist writing audio study material for students who learn better by listening, including students with visual impairment or dyslexia. Turn the text below into a narration script of 2 to 3 minutes (about 300 to 400 words). Rules: calm, friendly tone, short sentences (under 15 words); spell out every formula and unit in words; say powers of ten in words (for example "six times ten to the power eighteen", "ten to the power minus three"); no tables, bullet symbols or special characters; do not add facts that are not in the text; keep all numbers exactly as given; end with a 3-point recap. Text: [final corrected text pasted here]

NotebookLM version: upload the corrected text as a source, then paste only the rules and say "from the uploaded source". If it replies with an offer instead of the script, send: "Write the full script now."

What I learned: spelling numbers out in the prompt gave correct audio in all three tools. NotebookLM first answered with an offer to make an Audio Overview and only produced a script on the second attempt.

P3: Mind-map outline

You are an accessibility specialist making visual study aids for students who find dense text hard to read. From the text below, create a mind-map outline. Rules: one central idea; 5 to 6 branches, 2 to 3

short points per branch; each point under 8 words; keep all formulas, units and numbers exactly as given; write powers of ten with superscripts, for example 10^{-3}; do not add facts that are not in the text; plain text only, no tables or LaTeX. Text: [final corrected text pasted here]

What I learned: short points forced tools to drop detail ("nearly", the time in the example, units). Branch rules were followed best by Gemini and NotebookLM; ChatGPT used a four-point branch and repeated ideas.

P3b and A1: other prompts and settings

Item	Details
Mind-map picture prompt	[Add the exact prompt or instructions you gave the image/design tool, and the tool name.]
Follow-up fixes asked	Remove the garbled "current flow" label; add "AI-generated" and the name in the footer; remove the unlabelled resistor; draw the switch closed.
ElevenLabs settings	Free tier; model elevenlabs-v4; style "warmly"; final merged script pasted without titles or markers; output MP3, 149.9 seconds.

PART B: VERIFICATION REPORT

Scope: every fact, formula, number and unit in the simplified text, audio script and mind-map was checked against the NCERT text (Class 10 Science, Chapter 11, Section 11.1, pages 171-172, Reprint 2025-26). Sections beyond 11.1 were not used for the final aids.

B1. Method

- Each tool output was compared with the NCERT text line by line.
- Errors were logged with the tool name, what the AI wrote, what NCERT says, the error type, severity and the fix.
- A merged version was written from the verified parts. All formulas were rewritten in plain text.
- The audio was listened to in full. The mind-map picture was checked against the outline and the text.
- The student reads the final versions against NCERT once more before submission.

B2. Key facts checked against NCERT

Fact in the final aids	NCERT source	Status
Electric circuit = continuous, closed path for current; the bulb does not glow if it breaks	Sec 11.1, p. 171	Matches
Current = rate of flow of charge through a particular area in unit time	Sec 11.1, p. 171	Matches
Electrons were unknown at first; current taken opposite to electron flow	Sec 11.1, p. 171	Matches
I = Q/t (Eq. 11.1)	Sec 11.1, p. 172	Matches
Coulomb = charge in nearly 6×10^{18} electrons; electron charge is 1.6×10^{-19} C (negative)	Sec 11.1, p. 172	Matches
Ampere named after André-Marie Ampère (1775-1836); 1 A = 1 C / 1 s	Sec 11.1, p. 172	Matches
1 mA = 10^{-3} A; 1 μA = 10^{-6} A	Sec 11.1, p. 172	Matches
Ammeter measures current and is always connected in series	Sec 11.1, p. 172	Matches
Current flows from the positive to the negative terminal through the bulb and ammeter	Sec 11.1, p. 172	Matches
Example 11.1: 0.5 A × 600 s = 300 C	Example 11.1, p. 172	Matches

B3. Errors found in raw AI output, by type

Error type	Example from this project	Tool
Wrong or invented definition	Charge defined as "tiny charged particles moving through a wire"	ChatGPT
Added fact not in source	"Energy means the ability to do work"	ChatGPT
Wrong source attribution	Source tag "ALLEN's" on an NCERT definition	Gemini
Broken formula or exponent	10-3 with no superscript; 6×1018; V=IRV = IRV=IR	ChatGPT, Gemini

Error type	Example from this project	Tool
Lost precision	"Current increases when potential difference increases" instead of "directly proportional"; "nearly" dropped	Gemini
Conceptual slip	Electron called "a charge"; "current opposes electron flow"	NotebookLM, ChatGPT
Omission	Worked example, mA/μA, 1 A = 1 C/1 s, negative sign	All three, in some outputs
Tool artifacts	Citation numbers, LaTeX, leaked prompt text, an offer to make a podcast, an emoji	All three
Image error	Open switch beside a glowing bulb; garbled label	Image tool

B4. Responsible AI checks

Check	Result
Source is open and credited	NCERT Class 10 Science, Ch. 11, credited on the mind-map and in the reports
AI content labelled	Text, audio and picture are labelled AI-generated
No personal data uploaded	Only NCERT text, fictional data and anonymous peer responses
No voice cloning or deepfakes	Generic AI voice from ElevenLabs; no real person imitated
No copyrighted material copied in full	One short NCERT section used for study purposes, with credit
Free tools only	ChatGPT, Gemini, NotebookLM, ElevenLabs free tier, Canva or free design tool

B5. Remaining issues and limits

- One known error remains in the mind-map picture (switch drawn open while the bulb glows) until it is redrawn as closed.
- Peer feedback has not been collected yet, so clarity and usefulness are not verified by real reviewers.
- Checking was done on one section only. AI outputs can differ on each run, so a different run could produce different errors.
- AI-assisted checking is not a substitute for the student's own reading of the final versions against NCERT.

Student sign-off	Verified by: _____ Date: _____